Arish Shahab

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EDUCATION

McMaster University **Hamilton, Ontario, Canada**
Software and Biomedical Engineering (Bachelor of Engineering) | 4.0/4.0 GPA *Expected Graduation: May 2028*

Relevant Coursework: Linear Algebra, Data Science, AI & ML Technologies, Biomedical Innovation & Entrepreneurship

Honors/Awards: 1st Place - iBioMed Hackathon, 1st Place - HealthcareHacks, 2nd Place - McMaster Engineering Competition, 13th Globally - Harvard University International Research Olympiad

Extracurriculars: Google DSC Code Team Member, iBioMed 1st Year Ambassador, Writer @ *The Frequency (*McMaster Journal)

WORK EXPERIENCE

MedAce.AI **Toronto, Ontario, Canada**
Founder & Full-Stack Developer *Aug. 2024 – Present*

- Led the full-stack development of a medical school interview simulation tool using Python, Django, and React, which provided personalized feedback through NLP models and improved user retention by 35%.
- Managed the full-stack development, ensuring seamless integration of IoT and embedded software functionalities, contributing to the tool's interactive voice-based interview capabilities.

Software Engineering Researcher **Oct. 2024 - Present**
Johns Hopkins University | NeuroData Lab *Maryland, Baltimore, USA*

- Engineered machine learning algorithms using Python and TensorFlow, increasing computational efficiency by 35% in neural data analysis across over 200 datasets.
- Researched and applied advanced prompt engineering techniques to optimize large language model performance, achieving a 15% faster response rate.
- Developed AI models for healthcare applications using Flask and React, improving predictive analytics accuracy by 22%.

Software Engineering Researcher **July. 2024 - Oct. 2024**
Hamilton Health Sciences *Hamilton, Ontario, Canada*

- Developed a real-time data analysis platform using Python and Flask, reducing data processing time by 40% in orthopedic surgeries.
- Automated clinical trial data collection using JavaScript, AWS Lambda, and DynamoDB, decreasing manual data entry errors by 20%.
- Designed an interactive dashboard using React and Node.js, improving clinician engagement with patient recovery statistics.
- Collaborated with cross-functional teams to integrate machine learning models into the healthcare pipeline, ensuring data processing.

Software Engineering Fellow **July. 2024 - Oct. 2024**
Headstarter AI *San Francisco Bay Area, California, USA*

- Developed over 5 API applications using Next.js, Python, and OpenAI, achieving 98% accuracy and serving over 1,000 users, leading to a 40% improvement in data processing speed.
- Implemented AI platform infrastructure using Flask and Docker, reducing backend processing time by 30%.
- Designed machine learning models for business applications, resulting in a 20% increase in operational efficiency.

PROJECTS

SupraScan – AI Early Detection of Progressive Supranuclear Palsy (PSP) | *Frameworks & Languages Used: Python, TensorFlow, PyTorch, OpenCV, Scikit-learn, AWS, SQL, Streamlit, FastAPI*

- Developed a 3D CNN model analyzing PET/MRI scans for early PSP detection, achieving 75% accuracy and reducing false positives.
- Built a scalable AI pipeline integrating AWS S3, SQL, and FastAPI, optimizing scan retrieval and inference speeds by 40%.
- Engineered predictive analytics & anomaly detection models, providing clinicians with real-time insights on disease progression and patient risk assessment.

Harvard Hackathon 2025 | *Frameworks & Languages Used: Python, React, Flask, TensorFlow, SQL, Azure*

- Developed an interactive map using Azure's AI services, showing global rare disease prevalence and identifying 15% more recruitment gaps.
- Created deep learning models with Azure Machine Learning, improving clinical trial accessibility by 20%.

FitBud | *Frameworks & Languages Used: Python, pandas, Flask,, React.js, RAG, Firebase, Arduino*

- Designed a motion tracking system with Azure and Random Forest Classifier, achieving 92% exercise classification accuracy.
- Enhanced feedback speed by 30% through real-time data processing using Flask and Azure services.
- Utilized Azure's AI capabilities to optimize workout routines, reducing form errors by 20%.

ADDITIONAL INFORMATION

Programming Languages: Python, Java, C++, HTML, React, MATLAB, R Studio, SQL
Tools & Technologies: Microsoft Azure, Azure Machine Learning, Azure Cognitive Services, TensorFlow, PyTorch, Docker, Flask, Firebase
Certifications: AWS Certified Cloud Practitioner, AWS Certified Machine Learning, Google Cloud Certification